L.D. COLLEGE OF ENGINEERING

Department of Instrumentation & Control Engineering

Subject: Microcontroller and Interfacing (BE04000021)

LPG Leakage Detector Using 8051 Microcontroller

Microcontroller-Based Multi-Level Alerting, Ventilation Control & Safety Architecture

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1. Problem Statement

The LPG Safety Challenge

Liquefied Petroleum Gas (LPG) poses a significant danger due to undetected gas leakage, which can lead to devastating fires or explosions. In residential kitchens, the risk is increased by factors like aging hoses, user oversight, and poor ventilation.

Alarming Statistics

- Over 4,000 LPG-related incidents are reported annually in Asia alone.
- Faulty equipment (45%), human error (35%), and infrastructure failure (20%) are primary contributors to residential leaks.
- 70% of incidents occur in kitchens, with peak times during cooking hours.

The Need for an Automated Solution

A control system that detects LPG concentrations below the lower explosive limit, responds within seconds, provides clear warnings, and activates exhaust ventilation is required. Such a system must be affordable, easy to install, and robust enough to function reliably in cooking environments.

2. Existing Solution

Categories of Existing Solutions

Category	Examples	Strengths	Critical Limitations
Olfactory (Human-based)	Natural odour	Zero cost; no equipment needed	Olfactory fatigue, age-related decline
Basic Electronic Detectors	MQ-2/MQ-6 alarms	Low cost; fast response	No quantitative display; manual reset
Industrial/Professional Systems	Catalytic bead sensors	High accuracy; data logging	Expensive (>$500); complex installation

Specific System Requirements for the Proposed Design

- Detection threshold: Adjustable, factory-set at 1000 ppm
- Response time: < 3 seconds from gas presence to alarm activation
- Display resolution: 10-step bargraph or ppm reading on 16×2 LCD
- Alarm sequence: Warning (yellow LED + intermittent beep) at 1000 ppm; Critical (red LED + continuous beep) at 2500 ppm

Use-Case Scenarios

- Residential kitchens
- Small restaurants & food stalls
- LPG-fired water heaters in bathrooms
- Shared cylinder storage areas

3. Proposed Embedded System

Key Components:

- 8051 microcontroller with Flash memory
- MQ-2 gas sensor for LPG detection
- ADC0804 analog-to-digital converter
- 16×2 LCD display for real-time concentration reading
- Three LED indicators (green, yellow, red) and a buzzer for alerts

System Architecture:

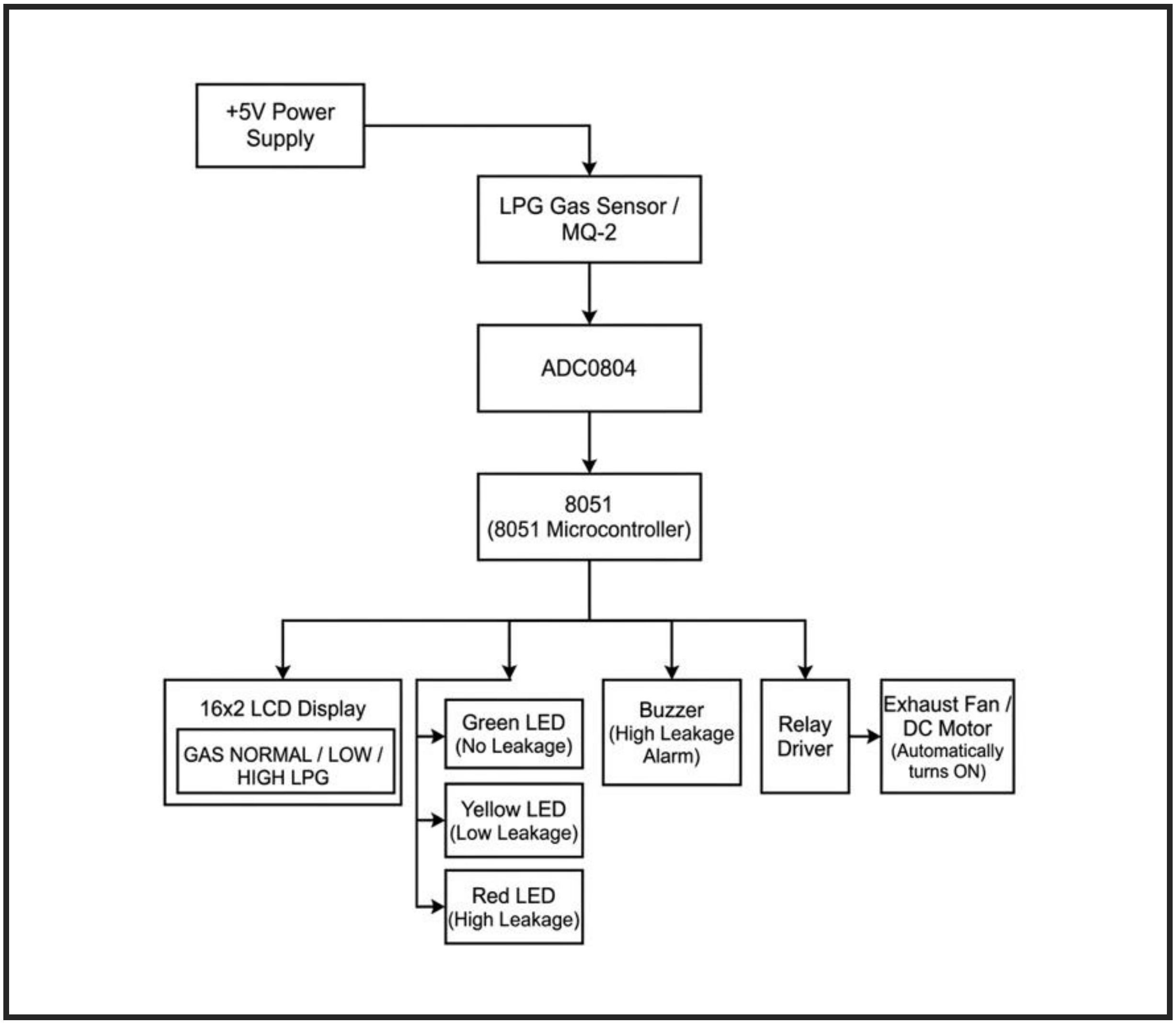
The 8051 microcontroller processes the digital value from the ADC and determines the appropriate response based on the gas concentration. The system includes a graduated response mechanism with three alert levels: SAFE, WARNING, and CRITICAL.

Safety Features:

1. Relay isolation via transistor driver
2. Fail-safe behavior through software reset routine
3. Sensor failure detection

4. Block Diagram

The diagram is structured to clearly separate the sensing, processing, and display, making the system easy to conceptualize and debug



5. Components

The proposed LPG leakage detector employs a carefully selected set of electronic components, each chosen for its specific function, reliability, cost-effectiveness, and compatibility with the overall system design.

Component Summary Table:

Component	Quantity	Function
8051	1	Main controller
MQ-2	1	LPG detection
ADC0804	1	Analog-to-digital conversion
LCD	1	Visual display
LEDs	3	Status indicators
Buzzer	1	Audible alert
Relay	1	Fan switching
DC Motor	1	Exhaust ventilation
Transistors	2	Drivers
Diode	1	Flyback protection
Resistors	8	Current limiting, pull-up, and load
Potentiometers	2	Contrast & calibration
Crystal	1	Clock source
Capacitors	3	Oscillator loading and decoupling
Supply	1	System power
Ground	1	Common reference

6. Sensor Selection:

The selection of an appropriate gas sensor is a critical design decision in any gas detection system. For this project, the MQ-2 gas sensor was selected after careful evaluation of multiple alternatives against a comprehensive set of criteria.

Sensor Technology Overview

Semiconductor sensors operate on the principle of changes in electrical conductivity when the sensing material is exposed to target gases. The sensing element is typically a metal oxide, such as tin dioxide (SnO_2), which is heated to a high temperature (200-400°C) to promote chemical reactions with gas molecules. In clean air, oxygen molecules adsorb onto the sensor surface, trapping electrons and creating a potential barrier that results in high resistance.

When reducing gases (such as LPG) are present, they react with the adsorbed oxygen, releasing electrons and reducing the potential barrier, which in turn lowers the resistance. This resistance change is proportional to the gas concentration and can be measured as a voltage change using a simple voltage divider circuit. Semiconductor sensors offer several advantages: they are low-cost, have fast response times (10-30 seconds), long operational life (up to 10 years), and provide analog output that can be easily interfaced with microcontrollers.

7. Actuator Selection:

Buzzer & Transistor Selection:

A piezoelectric buzzer was chosen for its high sound level, low power consumption, and ease of driving from a microcontroller port pin. The BC547 transistor is used as the driver due to its high current gain and low saturation voltage.

LED Indicator Selection:

High-brightness LEDs were selected for their clear visibility under typical lighting conditions. Each LED is connected through a 330Ω current-limiting resistor to ensure safe operation for both the LED and the microcontroller port pin.

Exhaust Fan (DC Motor) Selection:

A standard 12V, 120mm computer case fan was chosen due to its high airflow rating, low noise level, and compact size. The relay serves as the interface between the low-power microcontroller and the high-power exhaust fan.

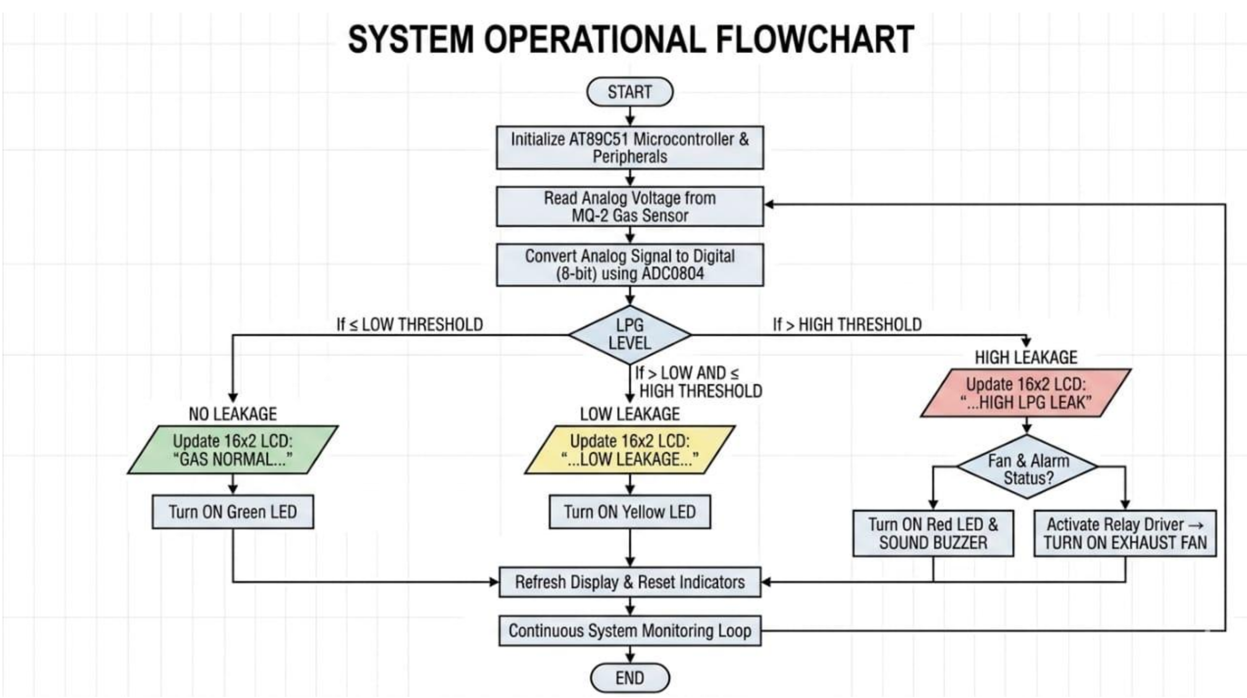
Relay Selection:

A 5V SPST relay with a contact rating of 5A at 12V was selected for its isolation properties and sufficient current handling capability. The BC547 transistor is used to drive the relay coil due to its high current gain and low saturation voltage.

Actuator Summary Table:

Actuator	Type	Quantity	Drive Circuit	Selection Rationale
Buzzer	Piezoelectric	1	BC547 transistor	Low power, high sound level
Green LED	High-brightness	1	330Ω resistor	Power/status indication
Yellow LED	High-brightness	1	330Ω resistor	Caution indication
Red LED	High-brightness	1	330Ω resistor	Danger indication
Exhaust Fan	12V DC, 120mm	1	Relay + BC547	Adequate airflow, low noise
Relay	5V SPST, 5A	1	BC547 + 1N4007	Isolation, contact rating
BC547	NPN transistor	2	1kΩ base resistor	Current gain, availability
1N4007	Rectifier diode	1	Parallel across relay coil	Back EMF protection

8. Flowchart



9. Algorithm

The algorithm implemented in the 8051 assembly code follows a continuous monitoring and response architecture. After initializing the microcontroller ports, ADC, and LCD display, the program enters an infinite loop that reads the 8-bit digital value from the ADC0804 (representing gas concentration) and compares it against two predefined thresholds (101 and 181). Based on this comparison, the system transitions into one of three states—NORMAL, LOW LEAKAGE, or HIGH LEAKAGE; each with a specific set of actions for LEDs, buzzer, exhaust fan, and LCD display.

Algorithm Steps :

```
1.  Program begins at address 0000H and jumps to MAIN routine.

2.  Configure Port 1 as input (ADC data lines) by writing 0FFH and Port 2 as output
    (LCD data lines) by writing 00H.

3.  Initialize all indicators: turn off green, yellow, and red LEDs (SETB); turn off
    buzzer and fan (CLR).

4.  Start ADC conversion by pulsing P3.2 low then high to generate WR signal.

5.  Call LCD_INIT subroutine to initialize the 16×2 display in 8-bit mode with 2 line
    s.

6.  Enter MAIN_LOOP: read the 8-bit ADC value from Port 1 into accumulator.

7.  Compare ADC value against threshold 101 (decimal). If less than 101, call NORMAL
    routine; else continue.

8.  Compare ADC value against threshold 181 (decimal). If less than 181, call LOW_LEA
    KAGE routine; else call HIGH_LEAKAGE routine.

9.  In NORMAL state: turn green ON, yellow and red OFF, buzzer and fan OFF; display "
    LPG GAS DETECTOR" and "GAS NORMAL" on LCD.

10. In LOW LEAKAGE state: turn yellow ON, green and red OFF, buzzer and fan OFF; disp
    lay "LPG GAS DETECTOR" and "LOW LEAKAGE" on LCD.

11. In HIGH LEAKAGE state: turn red ON, green and yellow OFF, buzzer and fan ON; disp
    lay "!!! LPG LEAK !!!" and "FAN + BUZZER ON" on LCD.

12. Each state routine updates LCD using LCD_CLEAR, LCD_CMD, and LCD_STRING subroutin
    es, then returns to MAIN_LOOP.

13. LCD subroutines handle command and data writes: LCD_CMD sends commands via Port 2
     with RS=0; LCD_DATA sends characters with RS=1; LCD_STRING reads from lookup tab
    le using DPTR.

14. DELAY_SHORT and DELAY_LONG subroutines provide timing using nested loop counters
    for proper LCD operation and loop pacing.

15. The RET instruction in HIGH_LEAKAGE returns to the main loop, while NORMAL and LO
    W_LEAKAGE jump directly back to MAIN_LOOP after their delay.

16. The infinite loop continues, continuously monitoring gas concentration and updati
    ng system outputs accordingly.
```

10. 8051 Pin Configuration

The 8051 microcontroller features 40 pins arranged in a Dual Inline Package (DIP), providing the interface between the microcontroller and the external world. Understanding the pin configuration is essential for proper circuit design and troubleshooting.

Pin Categories

The 40 pins can be divided into four functional categories: power supply, oscillator, control, and input/output (I/O). The I/O pins are further organized into four 8-bit ports (P0, P1, P2, P3).

Power Supply Pins: Pin 40 (Vcc), Pin 20 (GND)

Oscillator Pins: Pin 18 (XTAL1), Pin 19 (XTAL2)

Control Pins: Pin 9 (RST), Pin 29 (PSEN), Pin 30 (ALE/PROG), Pin 31 (EA/VPP)Input/Output Ports

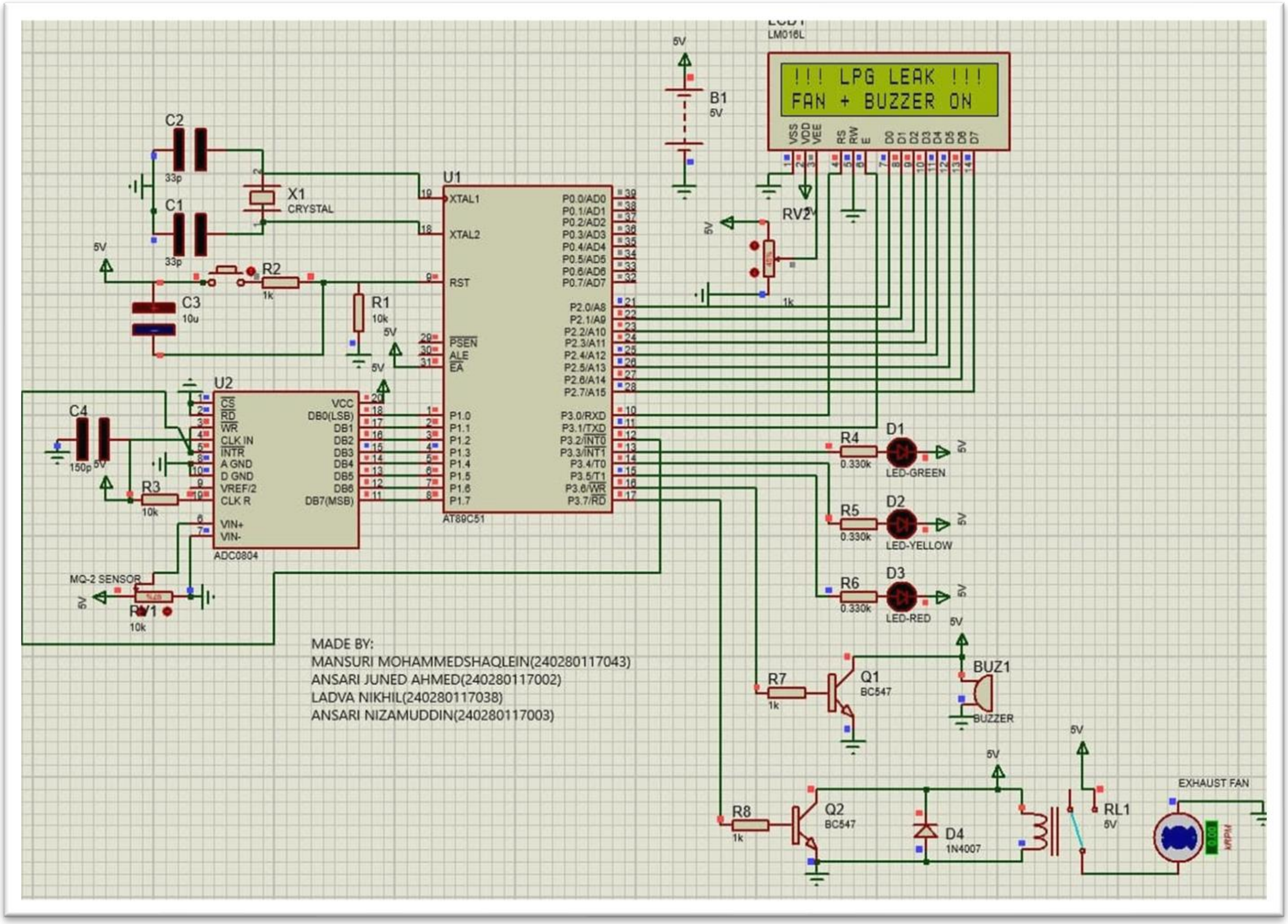
I/O Ports: Port 0 (Pins 32-39), Port 1 (Pins 1-8), Port 2 (Pins 21-28), Port 3 (Pins 10-17)

Pin Configuration Table

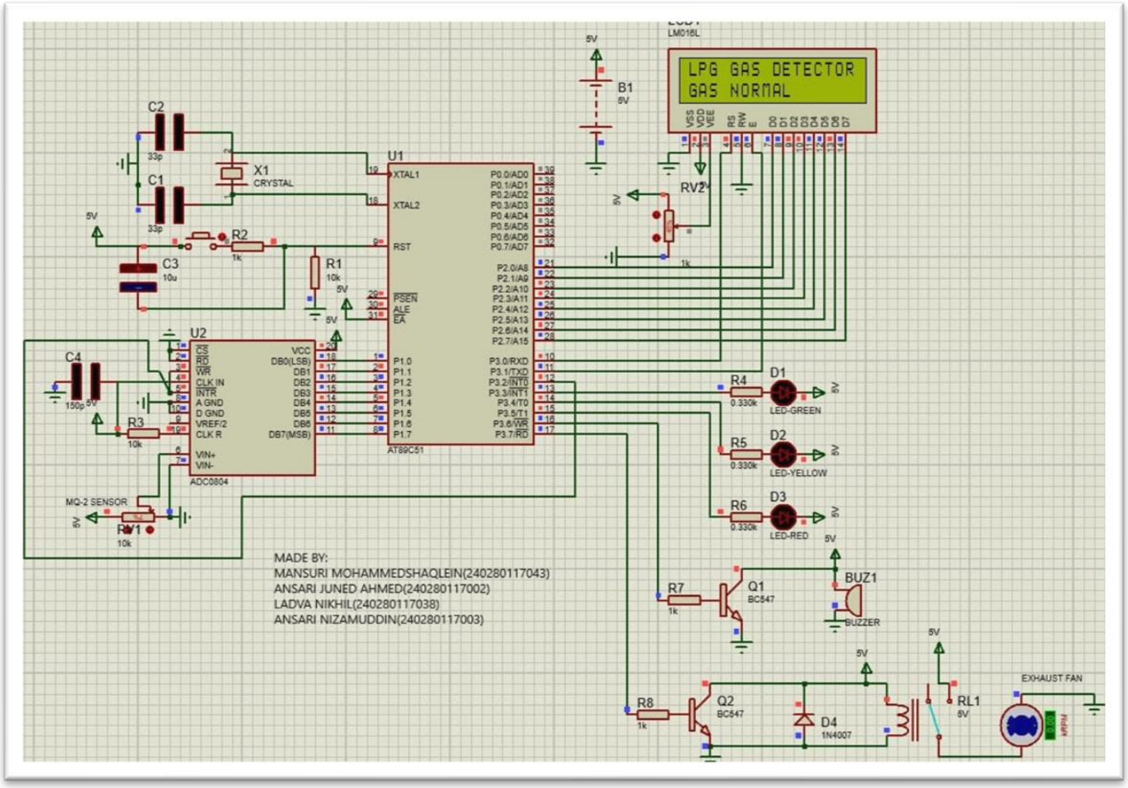
Pin	Name	Function	Connection
1	P1.0	LCD D0	LCD D0
2	P1.1	LCD D1	LCD D1
3	P1.2	LCD D2	LCD D2
4	P1.3	LCD D3	LCD D3
5	P1.4	LCD D4	LCD D4
6	P1.5	LCD D5	LCD D5
7	P1.6	LCD D6	LCD D6
8	P1.7	LCD D7	LCD D7
9	RST	Reset	Reset circuit
10	P3.0	LCD RS	LCD RS
11	P3.1	LCD RW	GND
12	P3.2	LCD EN	LCD EN
13	P3.3	Green LED	Green LED (330Ω)
14	P3.4	Yellow LED	Yellow LED (330Ω)

Pin	Name	Function	Connection
15	P3.5	Red LED	Red LED (330Ω)
16	P3.6	Buzzer	Buzzer driver
17	P3.7	Relay	Relay driver
18	XTAL1	Crystal	11.0592 MHz crystal
19	XTAL2	Crystal	11.0592 MHz crystal
20	GND	Ground	System ground
21	P2.0	ADC WR	ADC0804 WR
22	P2.1	ADC RD	ADC0804 RD
23	P2.2	ADC CS	ADC0804 CS (GND)
24	P2.3	ADC INTR	ADC0804 INTR
25-28	P2.4-P2.7	Not used	Unconnected
29	PSEN	Not used	Unconnected
30	ALE	Not used	Unconnected
31	EA	External Access	Vcc
32-39	P0.0-P0.7	ADC data	ADC0804 DB0-DB7
40	Vcc	Power	+5V

11. Interfacing Diagram



ADC value	Condition	Output
0–100	🟢 No leakage / Normal	Green ON, buzzer & fan OFF
101–180	🟡 Low leakage	Yellow ON, buzzer & fan OFF
181–255	🔴 High leakage	Red ON, buzzer ON, fan ON


LM016L
LPG GAS DETECTOR
GAS NORMAL
U1
AT89C51
U2
ADC0804
MQ-2 SENSOR
X1
CRYSTAL
LED-GREEN
LED-YELLOW
LED-RED
BUZ1
BUZZER
EXHAUST FAN
RL1
BC547
1N4007
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LADVA NIKHIL(240280117038)
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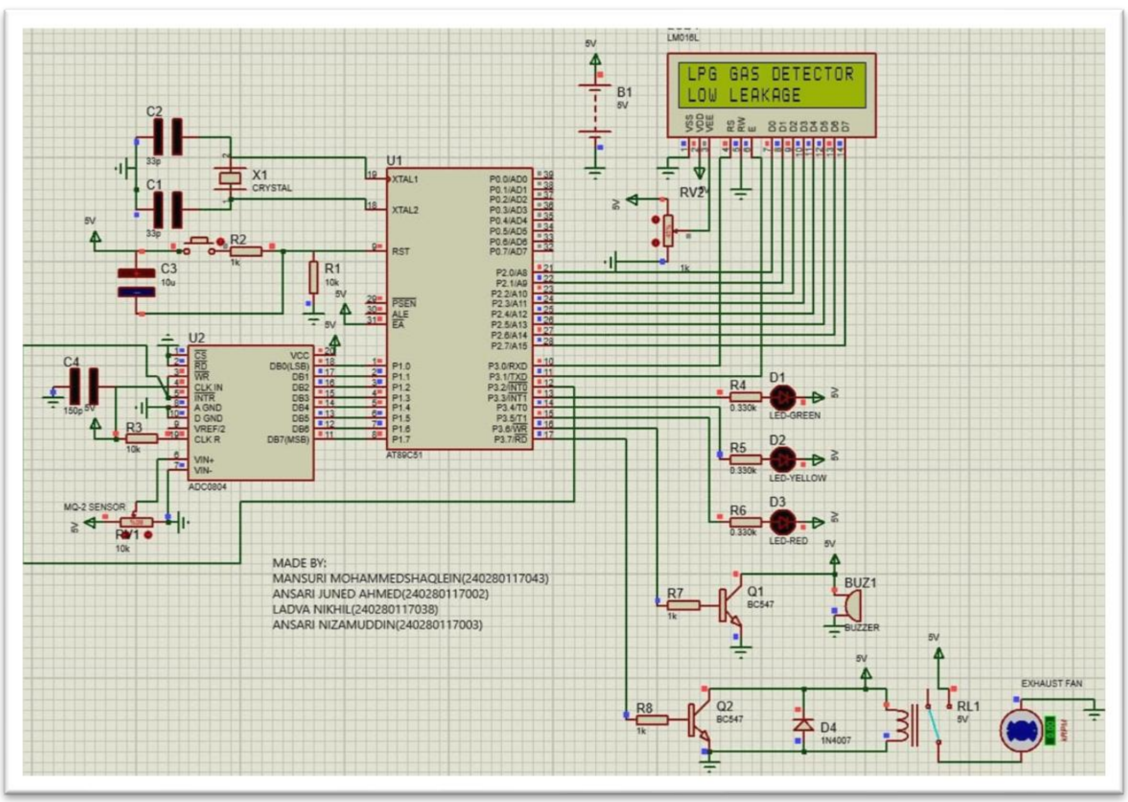
LM016L
LPG GAS DETECTOR
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12. Advantages

The proposed LPG leakage detector offers numerous advantages over existing solutions, combining the benefits of modern embedded systems with practical considerations for real-world deployment.

Advantages of the LPG Leakage Detector System

1. **Comprehensive Protection:** The system provides layered protection through multiple alert mechanisms and active mitigation for enhanced safety.
2. **Real-Time Continuous Monitoring:** The system continuously monitors gas concentration with rapid response, operating 24/7 without interruption.
3. **Quantitative Feedback:** The system provides real-time gas concentration readings in parts per million on the LCD display for accurate tracking.
4. **Cost-Effectiveness:** The system is designed using readily available and low-cost components, making it accessible to residential and small commercial applications.
5. **Ease of Implementation and Assembly:** The system uses standard components with well-documented interfaces and through-hole packaging, simplifying assembly and deployment.
6. **Low Power Consumption:** The system operates from a 5V supply and consumes minimal power under normal operation, allowing for battery backup or portable use.
7. **Expandability and Future-Proofing:** The AT89C51 microcontroller has unused I/O pins and built-in serial communication capability for easy expansion and future upgrades.
8. **Reliability and Longevity:** The system uses solid-state components with long lifespans, ensuring reliable operation over many years.
9. **User-Friendly Interface;** The system's visual and audible alerts are accessible to all users, regardless of hearing or visual ability.
10. **Autonomous Operation:** The system operates autonomously 24/7 without user intervention, providing continuous protection even when occupants are asleep or away.

13. Limitations

While the proposed LPG leakage detector system offers significant advantages and represents a meaningful improvement over existing solutions, it is important to acknowledge its limitations. Understanding these limitations is essential for proper deployment, realistic expectations, and future improvement efforts. No system is perfect, and recognizing the constraints of the current design allows users and developers to work within its boundaries and plan for enhancements.

Limitations of the LPG Leakage Detector System

1. **Sensor Selectivity and Cross-Sensitivity:** The MQ-2 sensor responds to various combustible gases and humidity changes.
2. **Sensor Calibration Requirements:** Regular calibration is necessary for accurate concentration readings.
3. **Warm-Up Time and Initial Instability:** The sensor requires a preheating period for stable operation.
4. **Limited Detection Range:** The sensor can detect concentrations between 300-10,000 ppm.
5. **No Gas Shutoff Capability:** Automatic gas shutoff is not included in the current design.
6. **Dependence on External Power Supply:** The system requires a continuous 5V DC power supply to operate.
7. **Single Point Monitoring:** The system uses a single sensor and may not detect leaks in areas not covered by the sensor.
8. **No Remote Notification or Monitoring:** The system does not include remote notification capabilities.
9. **Maintenance Requirements:** Regular cleaning and testing are necessary to ensure reliable operation.
10. **False Alarm Potential:** The system may trigger false alarms due to various factors such as cooking fumes or aerosol sprays.
11. **Limited Harsh Environment Suitability:** The system is designed for indoor use in typical residential and commercial environments.
12. **No Self-Test Capability:** The system does not include a self-test feature to verify its functionality.

14. Applications

The system is a cost-effective safety solution for any environment where LPG is used or stored. Its real-time monitoring, graduated alerts, and automatic exhaust activation make it suitable for diverse settings, from homes to small commercial facilities.

- Residential kitchens and cylinder storage areas
- Commercial restaurant and hotel kitchens
- LPG cylinder rooms and storage depots
- Small industrial workshops and factories
- Laboratories and research facilities using LPG

15. Future Improvements

The proposed LPG leakage detector system provides a solid foundation for gas leak detection and mitigation, but numerous opportunities exist for enhancement and expansion. These future improvements could transform the basic system into a comprehensive safety solution, addressing many of the limitations identified in the previous section while adding new capabilities that increase utility, reliability, and user convenience.

Feature	Complexity	Benefits
IoT Integration	Medium-High	Remote monitoring, alerts
Gas Shutoff Valve	Medium	Automatic shut-off in case of leak
Multiple Sensor Support	Medium	Extended coverage for larger spaces
Temperature and Humidity Compensation	Low-Medium	Improved accuracy, reduced false alarms
Data Logging and Analysis	High	Valuable for troubleshooting and compliance
Self-Test and Diagnostics	Medium	Improved reliability and user confidence
Improved User Interface	High	Enhanced user experience with graphs and touchscreens
Configurable Thresholds	Low	Customization for different environments
Battery Backup	High	Uninterrupted protection during power failures
Machine Learning and Pattern Recognition	Very High	Reduced false alarms, improved detection
Voice Alerts and Speech Synthesis	Medium	Clearer alerts, improved accessibility
Smartphone Application	Very High	Enhanced user engagement and remote control
Integration with Fire Safety Systems	High	Enhanced overall safety
Firmware Over-the-Air (OTA) Updates	High	Easy maintenance and upgrades
Improved Sensor Selection	Medium	Improved performance and flexibility

16. Estimated Cost

The estimated cost of the LPG leakage detector system is provided below, based on approximate current prices for components in single-unit quantities. These prices are indicative and may vary depending on supplier, quantity, region, and currency fluctuations. The cost breakdown includes all major components, passive parts, the PCB, enclosure, and power supply, providing a realistic total bill of materials (BOM) for building one complete unit. Cost optimization strategies are also discussed for bulk production or budget-conscious builders.

Component Cost Breakdown

Component	Quantity	Unit Price (₹)	Total (₹)
AT89C51 Microcontroller	1	70	70
MQ-2 Gas Sensor Module	1	90	90
ADC0804 IC	1	225	225
22516×2 LCD Display (with backlight)	1	150	150
Green LED (5mm, high-brightness)	1	2	2
Yellow LED (5mm, high-brightness)	1	2	2
Red LED (5mm, high-brightness)	1	2	2
Buzzer (5V, piezoelectric, 85dB)	1	15	15
5V Relay (SPST, 5A contact rating)	1	25	25
DC Motor / Exhaust Fan (12V, 80mm, 60CFM)	1	250	250
BC547 NPN Transistor	2	2	4
1N4007 Rectifier Diode	1	2	2
330Ω Resistor (1/4W, 5%)	3	1	3
1kΩ Resistor (1/4W, 5%)	2	1	2
10kΩ Resistor (1/4W, 5%)	2	1	2
10kΩ Potentiometer (single-turn, trimpot)	2	30	60
11.0592 MHz Crystal (HC-49)	1	10	10
33pF Ceramic Capacitor	2	2	4
10μF Electrolytic Capacitor (16V)	1	5	5
0.1μF Ceramic Capacitor (decoupling)	2	2	4
5V DC Power Supply (1A, wall adapter)	1	100	100
PCB (custom, single-sided, FR4, ~100×80mm)	1	300	300

Component	Quantity	Unit Price (₹)	Total (₹)
Enclosure / Case (plastic, with mounting holes)	1	200	200
IC Sockets (40-pin DIP, 20-pin DIP, etc.)	3	5	15
Headers, wire, solder, mounting hardware	-	20	20
Total Estimated Cost (Single Unit)			1565

Cost Optimization Strategies

1. *Volume Purchasing:* Buy components in bulk (30-50% cost reduction).
2. *Alternative Power Supply:* Use a USB power adapter or build a 5V power supply (~₹50 to ₹100).
3. *PCB Fabrication:* Order PCBs from budget fabrication services in small panels (~₹200 per unit).
4. *Enclosure:* Use a simple 3D-printed enclosure, repurpose an existing plastic box, or use a metal junction box (~₹50 to ₹150).
5. *Component Substitutions:* Replace components with lower-cost or surplus alternatives (e.g., generic buzzer for ₹15).
6. *IC Sockets:* Omit IC sockets and solder ICs directly to the PCB (save ~₹15-20).
7. *DIY Sensor Module:* Use a bare MQ-2 sensor element (~₹40 to ₹60) instead of a pre-made module.

17. Innovation

The LPG leakage detector system introduces practical innovations that transform conventional gas detection into an active safety solution. By integrating real-time monitoring, graduated alerting, and automatic ventilation into a unified, low-cost platform, the system addresses critical gaps in existing residential and small commercial safety products.

Unlike basic detectors that merely sound an alarm, this system actively mitigates hazards by automatically activating an exhaust fan upon detecting dangerous gas concentrations. The graduated alert system with software hysteresis ensures stable, chatter-free operation while providing early warning for minor leaks. Quantitative concentration display on a standard LCD empowers users with actionable information, bridging the gap between simple alarms and industrial-grade systems.

Key Innovations

1. **Unified Safety Response:** Seamlessly integrates detection, graduated alerting, and active ventilation into a single autonomous system that both warns occupants and works to eliminate the hazard.
2. **Intelligent Graduated Alerting with Hysteresis:** Three-level alert system (CAUTION and DANGER) with software hysteresis prevents nuisance chattering and provides early warning before critical escalation.
3. **Quantitative Visualization at Low Cost:** Real-time concentration display in ppm on a standard LCD empowers users with actionable data, transforming the device from a simple alarm into a diagnostic tool.
4. **Affordable Active Mitigation:** Brings exhaust fan control to the residential market at under $50, making proactive gas safety accessible to households and small businesses.
5. **State-Machine-Based Software Design:** Deterministic state machine with hysteresis-driven transitions ensures predictable, reliable operation under all conditions.
6. **Strategic Component Selection:** Uses widely available, industry-standard parts to achieve supply chain resilience, long-term maintainability, and ease of reproduction.
7. **Open-Source and Community-Ready Architecture:** Modular design with well-documented interfaces encourages peer review, knowledge sharing, and continuous improvement.
8. **Practical Safety Impact:** Directly addresses the persistent public safety challenge of LPG-related fires and explosions with a deployable, low-cost solution for widespread adoption.

18. Conclusion

This report has presented a comprehensive design and implementation of an LPG leakage detector system built around the AT89C51 microcontroller. The system effectively addresses the critical safety challenge of LPG leaks through a practical combination of real-time monitoring, graduated visual and audible alerts, and automatic mitigation via an exhaust fan. The integration of an MQ-2 gas sensor, ADC0804, and 16×2 LCD display provides reliable detection with quantitative feedback at a low cost.

The project demonstrates that sophisticated safety features—including hysteresis-controlled thresholds, state-machine-based decision making, and active ventilation—can be realized with standard, off-the-shelf components. The estimated cost of under $50 makes this solution accessible for residential and small commercial applications, bridging the gap between basic commercial detectors and expensive industrial systems.

The system's modular design and use of widely available parts ensure ease of assembly, maintenance, and future expansion. While certain limitations exist, such as sensor cross-sensitivity and dependence on external power, these can be mitigated through careful deployment and potential enhancements like remote monitoring and gas shutoff integration.

Conclusion: *The proposed LPG leakage detector is a practical, innovative, and cost-effective safety solution that can significantly reduce the risk of gas-related accidents. Its adoption has the potential to save lives and prevent property damage, making it a meaningful contribution to public safety and embedded systems application.*

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